A LiDAR-Enabled Ground Rover Framework for Smart Monitoring in Utility-Scale Solar Power Systems: Design and Preliminary Validation*

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Abstract

Smart monitoring and reliability are becoming increasingly important in utility scale photovoltaic (PV) systems as solar farms expand in size and play a larger role in sustainable power infrastructure. This paper presents a ground-based robotic inspection framework designed to support reliability-centered monitoring in utility-scale solar farms using a four-wheel-drive unmanned ground vehicle equipped with a Livox Mid-360 LiDAR sensor and an articulated robotic arm. The proposed system addresses a practical gap in current inspection practice: many structural and geometric anomalies, including panel displacement, frame deformation, vegetation encroachment, and terrain change, can emerge before they are clearly reflected in electrical output data.

The paper introduces the rationale for the design, the architecture of the system, and the proposed platform inspection workflow. Early laboratory trials at the Pacific Earthquake Engineering Research (PEER) Center's laboratory confirmed the feasibility of the LiDAR-based point cloud acquisition and the three-dimensional reconstruction process, directly informing a key design refinement via replacement of a fixed LiDAR mount with an articulated arm to improve scan coverage and reduce occlusion. The paper also outlines the planned validation pathway, including field deployment using a portable PV testbed with a string inverter. Overall, this work presents a practical framework for advancing data-driven, spatially informed, and reliability-centered operations and maintenance in utility-scale solar power systems, with field validation planned as the next step.

1 Introduction

Reliability is becoming a critical performance objective in utility-scale photovoltaic (PV) plants alongside energy production. As solar farms increase in size and continue to age,

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demands for operations and maintenance (O&M) are increasing and the consequences of undetected degradation are becoming more significant. Recent reviews of smart grid development consistently identify the observability and predictive maintenance of renewable assets as essential requirements for next generation power infrastructure [1, 2].

One of the main challenges in large PV farms is that many failure precursors begin as physical and spatial changes before they become electrical problems. Storm events, for example, can shift panel mounting angles or deform support frames. Vegetation can invade the rows of the array and soil can settle beneath ballasted racks. Typically, these conditions do not necessarily cause an immediate drop in power production, yet all of them represent an increasing risk to structural integrity and long term performance [4]. Electrical monitoring alone cannot detect such problems early enough for effective decision-making to take corrective measures. Manual visual inspections are difficult to carry out on the scale of modern solar farms. Although unmanned aerial vehicle (UAV)-based inspection has advanced considerably over the past decade, aerial platforms still have clear limitations and restrictive regulations for close geometric inspection tasks. They cannot reliably inspect beneath panels, measure small structural displacements with confidence, or consistently return to the same viewpoint for detailed change detection [6].

This paper addresses this gap by presenting the design and planned deployment of a LiDAR-equipped ground rover that can move through solar farm rows, repositioning its sensor using a robotic arm, and generating dense three-dimensional (3D) point clouds of panel surfaces, structural members, and the surrounding site conditions. The method is not intended to replace aerial inspection. Instead, it is proposed as a complementary ground level capability that can provide spatial information at a scale and in situations where drones and manual inspections cannot be consistently achieved.

The remainder of the paper is organized as follows. Section 2 reviews the state of PV inspection and identifies the specific monitoring gap. Section 3 describes the hardware platform and the design decisions that may influence its operation and performance, including lessons learned from early laboratory trials. Section 4 outlines the proposed inspection workflow. Section 5 discusses the intended applications and the expected benefits. Section 6 presents the current state of the work, its limitations, and the planned validation program. Section 7 concludes the paper.

The main contributions of this work are:

- A ground-based robotic inspection architecture that integrates LiDAR sensing with articulated sensor positioning;
- A structured workflow for repeatable geometric inspection and change detection in solar arrays; and
- A validation roadmap linking laboratory calibration, controlled testing, and field deployment.

2 Background and Related Work

2.1 O&M Challenges in Utility Scale PV Systems

The solar farm O&M has traditionally been reactive, with teams responding to alarms, inverter faults, or visible physical damage. As installations continue to expand on an ongoing basis, this approach becomes increasingly ineffective. A plant that covers several hundred acres, or even areas approaching ten square kilometers, such as the Abydos Solar Plant, Kom Ombo, Egypt or the Benban Solar Park project, western desert of Egypt, that extends to roughly 37 km^2, cannot be inspected efficiently by walking on the site. Under such conditions, faults localized to individual strings or modules may remain undetected for extended periods. Recent reviews of PV O&M practice show a clear shift towards *predictive and reliability-based maintenance*, where asset level condition monitoring plays a central role [7, 8]. These studies also highlight the practical limitations of manual inspection in large PV plants. Although field crews can perform a close visual assessment when physically present at a location, manual inspection is labor intensive, difficult to repeat with high positional consistency, and relatively slow when large areas must be covered systematically.

Environmental exposure makes this challenge even more difficult. PV systems are subject to dust accumulation, thermal cycling, humidity, wind fatigue, and occasional extreme events such as hail, strong winds, and flooding, all of which can produce gradual degradation or sudden structural damage [3]. The structural consequences are often not visible in the electrical monitoring data. A loose mounting bolt, a support member that has deformed after a wind event, or a panel that has shifted slightly out of alignment may not immediately affect the energy output, yet these conditions still matter for long term reliability.

2.2 Aerial Inspection: Strengths and Limitations

UAV-based inspection is now well established in large PV plants and has generated extensive research interest. Drone-based thermal imaging can identify hotspot anomalies, bypass diode failures, and soiling patterns in large arrays in a rapid and non-intrusive manner [9, 10]. Aerial monitoring supported with digital twins[1] has also shown promise for more systematic tracking of asset condition [5]. These developments demonstrate the particular strengths of UAV-based inspection in wide area coverage and rapid thermal assessment.

The limitations of aerial inspection are important to recognize. Drones are not ideal for close geometric measurement because depth resolution decreases as flight altitude increases

[1] A digital twin is a dynamic virtual representation of a physical system that integrates models with real-time or historical data to monitor, simulate, and predict system behavior over its lifecycle. In PV applications, digital twins enable performance assessment, fault detection, and predictive maintenance by combining physics-based models with operational data.

to maintain safe clearance and to comply with existing minimum height regulations. Repeated positioning within a few centimeters of the same location, which is necessary for consistent change detection, is difficult to guarantee under realistic wind conditions. In addition, visibility beneath panels and within rack structures is essentially absent from an overhead inspection view. For detecting small structural deformations or subtle geometric changes, these are not minor issues but fundamental limitations [9, 6].

2.3 Ground Robots for PV Inspection

Unmanned ground vehicles (UGVs) have attracted increasing attention as inspection tools because they complement aerial systems rather than compete with them. A 2025 review of UGVs for the inspection of PV plants found that ground robots are especially well suited for repeated path-based monitoring, close geometric sensing, and inspection tasks that require a stable and ground referenced viewpoint [6]. Unlike drones, UGVs are not affected by wind during sensing, can carry heavier sensor payloads, and can more reliably be programmed to revisit the same physical location.

LiDAR is a particularly suitable sensing technology for the monitoring application of PV farms using UGVs. It provides direct and metrically consistent 3D point clouds without dependence on ambient lighting conditions, and it can capture small geometric differences between repeated scans. In the context of solar farm inspection, this creates the possibility of detecting changes in panel tilt, frame displacement, reduced clearances, and vegetation growth through quantitative analysis rather than visual judgment alone. Table 1 summarizes the main capability differences between the proposed rover–LiDAR system, a conventional fixed-mount UGV configuration, and the principal inspection approaches discussed above.

2.4 Research Gap

Despite the growing body of work on PV inspection, there is still no clearly established framework that integrates a mobile ground robot, articulated LiDAR-based sensing, and repeat-inspection geometric change detection specifically targeting early-stage structural anomalies in utility-scale solar arrays. This paper proposes such a framework, emphasizing close-range geometric observability and repeatable spatial monitoring as complementary capabilities to existing aerial inspection methods.

3 System Architecture and Platform Design

3.1 Platform Selection

The mobile base selected for this work is the *ROVER ZERO* 4WD UGV. It was chosen for its rugged outdoor capability, Intel NUC i7 onboard computing, integrated motor control, and full ROS and ROS2 compatibility. The platform includes a payload deck

Table 1: Comparison of solar farm inspection approaches.

Criterion	**Manual**	**UAV**	**Fixed UGV**	**Proposed Method**
Close-range geometric inspection	△	✗	✓	✓
Under-panel and rack visibility	△	✗	△	✓
Repeatable positioning	✗	△	✓	✓
Occlusion[a] reduction via articulated arm	✗	✗	✗	✓
Thermal and hotspot detection	✗	✓	△	△
Large-area coverage speed	✗	✓	△	△
Low sensitivity to wind during sensing	✓	✗	✓	✓
No airspace approval required	✓	✗	✓	✓
Change detection over time	✗	△	△	✓
Relative operating cost	High	Medium	Medium	Medium

[a] Occlusion refers to the partial or complete blockage of an object from view by another object.
Symbols: ✓= supported; ✗= not supported; △= partially supported.

with sufficient area and capacity for the LiDAR, robotic arm, and associated cabling. Its four-wheel-drive configuration is also well suited for the uneven terrain commonly found at utility scale solar sites. The LiDAR sensing unit was installed on this rover platform, as illustrated in Fig. 1.

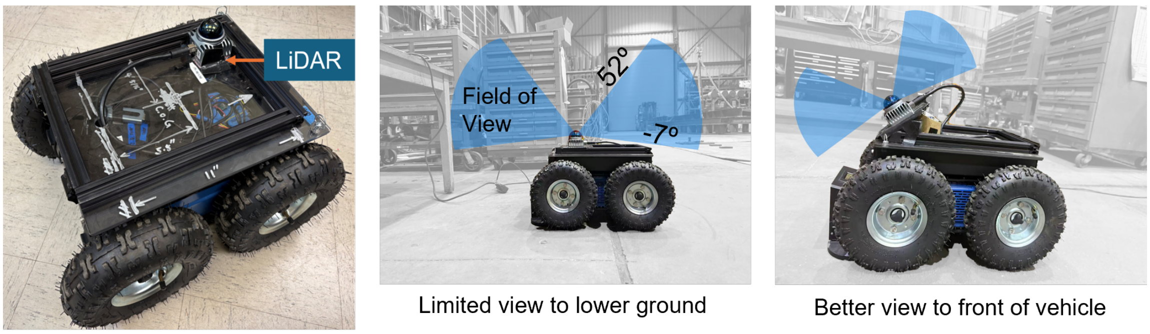

Figure 1: LiDAR sensing unit installed on the ROVER ZERO 4WD UGV platform.

3.2 LiDAR Scanning

The primary sensor is the *Livox Mid-360*, a solid state LiDAR with a 360° horizontal field of view, a 59° vertical field of view, a range of up to 70 m, and a point rate of approximately 200,000 points per second. The sensor is integrated with the rover's Intel NUC (Next Unit of Computing) and operates using the open source *FAST LIO2* simultaneous localization and mapping (SLAM) framework for real time pose estimation and map building. As illustrated in Fig. 1, the initial LiDAR installation provides a limited field of view near the lower ground area in front of the rover. However, an improved mounting angle enhances the scanner's forward visibility. Based on the initial observations and the improved visibility achieved in the revised mounting configuration, it became evident that a fixed LiDAR installation is insufficient for capturing all relevant inspection zones, especially beneath the PV modules' support beams. These results motivated the integration of a robotic arm for flexible and reliable sensing perspectives, as discussed further in Section 3.3.

Early laboratory trials were conducted at the PEER laboratory at the University of California, Berkeley, where the system was used to scan structural test frames under controlled conditions. These experiments were intended as an initial proof of concept to evaluate whether the LiDAR setup could generate reliable 3D reconstruction of complex scenes. As shown in Fig. 2, these trials confirmed that the LiDAR and SLAM stack can generate dense and well registered point clouds suitable for geometric inspection tasks, which provided the foundation for the planned adoption to utility-scale solar farms. They also revealed a consistent limitation in the fixed mount configuration. The regions located directly behind or beside the rover body created systematic occlusion zones that left gaps in the reconstructed geometry. This limitation was therefore attributed not to the sensing unit itself, but to the original mounting arrangement.

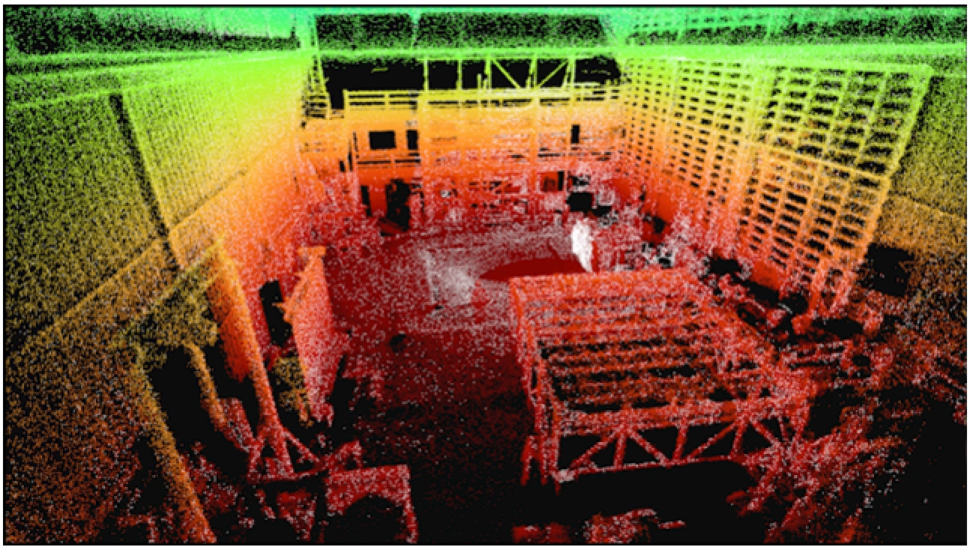

Figure 2: LiDAR point cloud of PEER laboratory space, colored by elevation (red = lower regions, green = higher regions), showing achieved scanning density and spatial coverage.

3.3 Articulated Arm for Adaptive Sensor Positioning

One of the most important design decisions in this project was replacing the fixed LiDAR mount with an articulated robotic arm after the occlusion issue was identified in early tests. The selected arm is the *Arduino Tinkerkit Braccio*, shown in Fig. 3, a six-axis servo actuated platform with a reach of up to 80 cm and a payload capacity of approximately 400 g, which is sufficient for the Livox Mid-360 sensor.

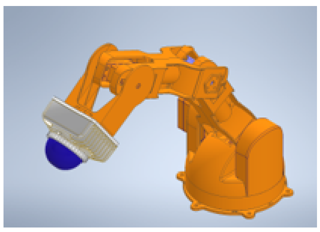

Figure 3: Robotic arm.

The arm is used as an adaptive sensor positioning mechanism. Its primary purpose is to elevate and orient the LiDAR so that the sensor can capture scans from a wider range of viewing angles and reduce the dead zones observed in the original fixed mount configuration. In practice, this allows the rover to stop at a designated inspection point and reposition the LiDAR through a sequence of preprogrammed poses that improve visibility around the rover body and the surrounding site features. By increasing the sensor height and adjusting orientation, the system is able to obtain more complete 360° scan coverage during inspection.

This design choice is particularly important for solar farm monitoring, where inspection quality depends on clear visibility of panel surfaces, support members, surrounding vegetation, and nearby geometric features. A fixed low mounted sensor can leave portions of the scene partially hidden behind the rover structure or outside the effective line of sight. Elevating the LiDAR on the articulated arm helps overcome this limitation and improves the completeness of the collected point clouds. This design introduces additional mechanical complexity and creates the need to evaluate the arm positioning accuracy in different payload and joint configurations. That assessment is part of the planned validation program described in Section 6.

3.4 Autonomous Navigation and Scanning Integration

In addition to sensing, the rover platform supports autonomous navigation and scanning using a ROS2-based pipeline, Fig. 4. LiDAR data are processed through the FAST-LIO2 framework for real-time pose estimation and scan registration, while a SLAM toolbox constructs a navigation map of the environment. The Nav2 stack is used for path planning, control, and waypoint-based navigation, allowing the rover to follow predefined inspection paths while continuously acquiring spatial data. This integrated pipeline enables coordinated motion and sensing, forming the basis for repeatable and automated inspection workflows.

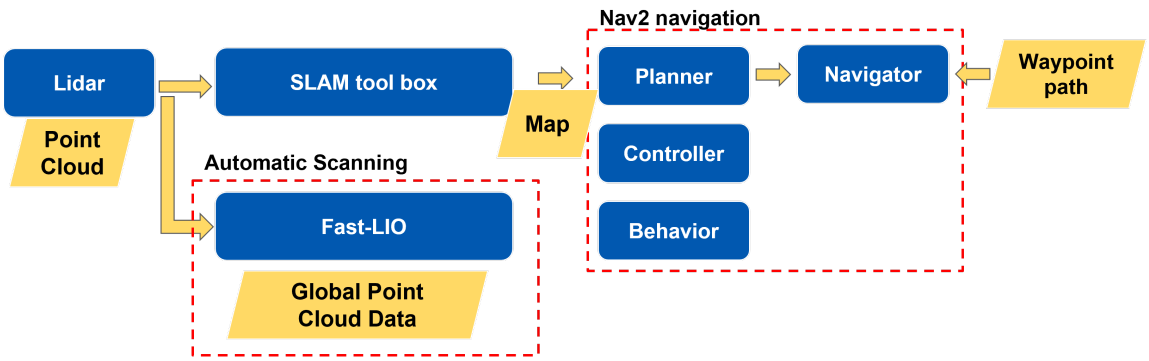

Figure 4: Pipeline for autonomous inspection integrating sensing (LiDAR), localization & mapping (FAST-LIO2 & SLAM), and navigation (Nav2) for waypoint-based scanning.

3.5 Reconstruction Pipeline

The raw point clouds from FAST LIO2 are processed using a *dual marching cubes algorithm*[2] implemented in Blender to generate reconstructed surface meshes of the inspected geometry. Early trials showed that reconstruction quality is sensitive to point cloud density and variance. Sparse regions can produce visible artifacts, and returns from reflective panel surfaces require *outlier filtering*[3] before reliable meshing is possible. More robust semantic segmentation and surface specific filtering are planned improvements to the pipeline.

3.6 Portable PV Testbed

Field evaluation will be conducted using a purpose-built portable PV test system consisting of a wheeled and ballasted frame that carries eight PV modules, a string inverter, and integrated cable management. The system operates from an onboard compact energy storage unit, removing the dependence on site power during field trials. The use of a string inverter provides an electrical architecture that is more representative of field deployed utility-scale PV systems and supports future integration between LiDAR-based geometric observations and system-level electrical performance monitoring. The testbed is assembled on wheels will rotate through multiple locations at the test site to expose the inspection system to varying shading profiles, surface conditions, wind exposure, and thermal environments while maintaining the electrical architecture unchanged.

Figure 5: Testbed schematic (left) and under construction (right).

[2]The dual marching cubes algorithm is a mesh extraction technique used to generate smooth polygonal surfaces from volumetric data. Unlike the classical marching cubes method, which places vertices on grid edges, the dual formulation constructs vertices within grid cells, improving mesh quality and preserving sharp features. In Blender (an open-source 3D computer graphics software used for modeling, rendering, animation, and simulation. It supports mesh processing, including remeshing and surface reconstruction, and is widely used in both graphics and engineering workflows), it is implemented (e.g., via remeshing tools) to convert implicit or voxelized geometry into clean, manifold surface meshes.

[3]Outlier filtering is a preprocessing step in which observations that lie far from the dominant data distribution are detected and removed using statistical or model-based criteria, thereby enhancing the reliability and stability of downstream estimation or learning tasks.

4 Proposed Inspection Methodology

The proposed inspection workflow consists of three main phases. Figure 6 provides an overview of the complete sequence, from rover deployment and continuous mapping through station-based scanning to geometric change detection and maintenance prioritization.

Phase 1 — Deployment and traversal. The rover follows a predefined path along or between panel rows at a nominal traversal speed selected to remain compatible with the tracking stability of the FAST LIO2 SLAM pipeline. In the current proof-of-concept configuration, preliminary operating speeds in the range of 0.2–0.4 m/s are being considered. The pipeline runs continuously throughout the traversal, building a localized map, and registering point clouds from successive rover positions into a single coordinate frame. In this phase, the objective is to establish the overall geometry of the inspection scene and to maintain a consistent spatial registration during the inspection route.

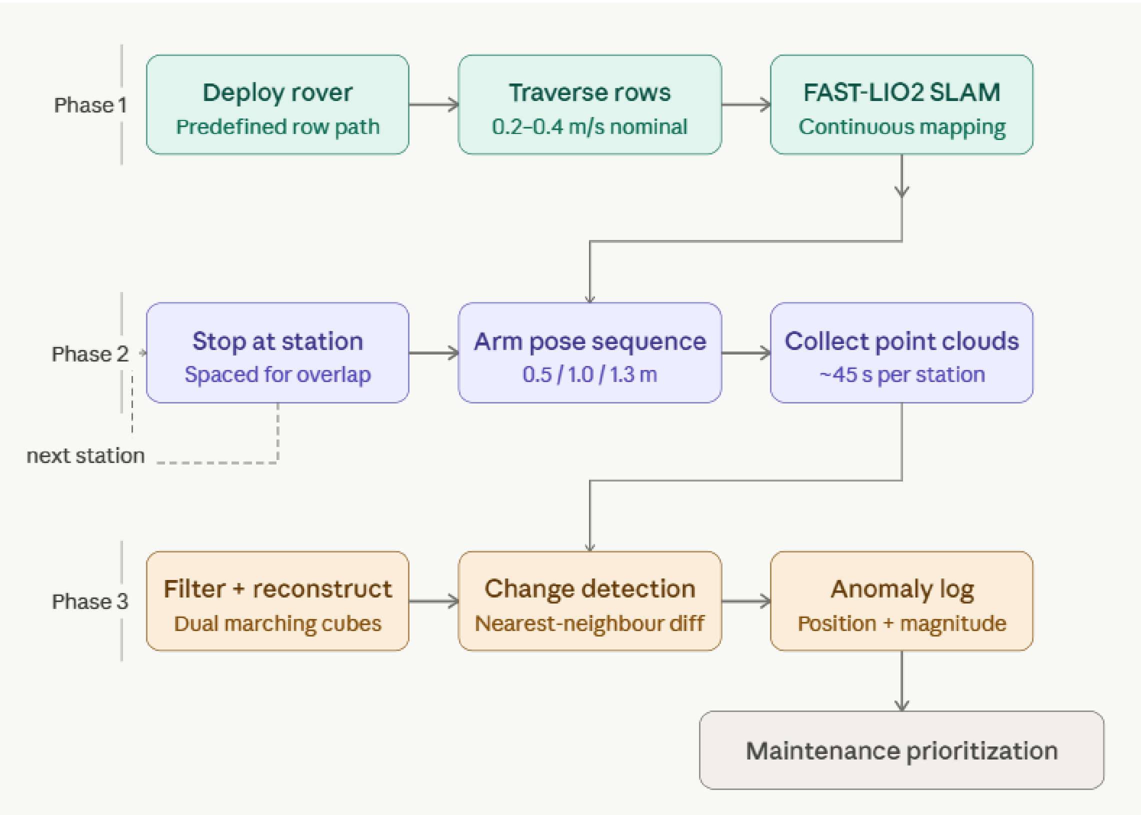

Figure 6: Proposed three-phase inspection workflow for the proposed system.

Phase 2 — Station-based scanning. At designated inspection sections, the rover stops, and the Braccio arm lifts the LiDAR to an elevated scanning position intended to reduce the occlusion zones identified during laboratory testing and improve scan coverage around the rover body and nearby array components. The arm is used primarily to raise the sensor and improve visibility. In the current setup, a complete station scan is expected to take approximately 45 seconds, although this value will be evaluated and refined during the validation efforts.

Phase 3 — Post-processing and change detection. Collected point clouds are processed through outlier filtering, surface normal estimation, and the dual marching cubes reconstruction pipeline. For repeated inspections, registered clouds from successive visits are compared using a nearest-neighbor distance metric to identify regions where measured geometric change exceeds a selected threshold. These regions are then flagged as candidate anomalies and recorded with their spatial position, magnitude of change, and scan timestamp to support maintenance prioritization. The specific thresholds used for anomaly detection will be determined during the calibration phase of the validation program, using ground-truth measurements from the PEER laboratory test setup as the primary reference. The performance of the change detection process will be evaluated in terms of detectable displacement thresholds, spatial registration error, and the rate of false positive and false negative anomaly identifications during validation.

5 Intended Applications and Expected Benefits

5.1 Primary Application: Early Anomaly Detection

The central application of the proposed system is to detect physical changes in solar farm arrays before they impact the electrical output. As noted in recent O&M reviews, the gap between when a structural issue develops and when it becomes visible in production data may persist long enough for the condition to progress before corrective action is taken [7, 8]. During that period, uncorrected conditions can progress from minor misalignments to more serious structural or safety hazards that require more costly intervention.

The system is designed to detect the following classes of anomaly: panel displacement or tilt change after high-wind or storm events; progressive frame deformation along mounting structures; vegetation encroachment near row clearances and cable trays; terrain settlement beneath ballasted rack foundations; and post-event structural changes that are not yet severe enough to reduce output by a detectable margin. Each of these conditions is difficult to resolve from electrical data streams or overhead imagery alone.

5.2 Secondary Application: Baseline Documentation

A second important application is the creation of dense, georeferenced point-cloud records of the *as-inspected* array, refer to [11, 12] for related earlier studies related on the use of laser scanning to determine the *as-found* construction configuration and geometry. Current documentation practice in large PV plants is heavily based on photographs, manual field notes, and occasional aerial imagery. These records are difficult to compare systematically over time and rarely provide the spatial resolution needed for precise structural evaluation.

Repeated LiDAR scans of the same site can serve as a metrically consistent baseline for accurate damage assessments following storm events, post-decommissioning condition surveys, and legal or warranty disputes involving structural performance. For example, a point-cloud record of a farm taken shortly after commissioning provides an objective reference against which future structural deviations can be measured quantitatively rather than estimated from photographs.

5.3 Expected Benefits

The expected benefits of the framework are both technical and operational. Technically, the system enables ground-level 3D sensing at a spatial resolution and repeatability that aerial platforms cannot consistently achieve for close-range geometric tasks. The combination of rover mobility and articulated arm positioning enables viewpoints that neither drones nor fixed sensors can access, including near-horizontal angles on panel undersides and direct views of rack-to-ground connections.

Operationally, earlier structural awareness results in better-informed maintenance scheduling. Rather than dispatch inspection crews reactively after a production anomaly appears, plant operators can use scheduled rover passes to triage the physical condition of the site and prioritize corrective work before failures spread. Recent reviews of PV O&M practice have documented that *reactive maintenance strategies* carry significantly higher per-fault costs than condition-based approaches, and that even incremental improvements in early detection lead-time have measurable economic impact at the scale of utility plants [7, 8]. The proposed platform is positioned to support this shift towards more proactive and data-driven O&M in large-scale solar infrastructure.

6 Current Status, Limitations, and Future Work

The rover has been procured, assembled, and tested. The LiDAR and SLAM stacks are operational. Laboratory scanning trials at PEER confirmed point cloud generation and led directly to the shift from a fixed LiDAR mount to an arm-mounted configuration. The Braccio arm has been mechanically integrated onto the rover payload deck, and initial bench level pose sequences have been programmed and verified. The design of the portable PV testbed has been completed, and procurement and assembly have been completed.

The system has not yet been deployed for inspection of an actual solar farm. No field scan dataset is available at this stage and the anomaly detection and change comparison pipeline has not yet been validated against known ground truth. Reconstruction quality in outdoor environments, where panel surfaces can create specular returns and environmental conditions vary, has not yet been fully characterized. The repeatability of the positioning of the robotic arm under field conditions, including changes in temperature and vibration during the movement of the rover, remains to be measured as well. These represent the primary open questions that define the scope of the validation program to be implemented.

The planned validation program consists of three phases. First, controlled laboratory scanning of a representative panel and frame assembly will be used to calibrate the change detection threshold and quantify reconstruction accuracy against a surveyed reference configuration. Second, shaking table tests at the PEER laboratory will allow LiDAR derived deformation estimates to be compared against ground truth displacements from accelerometers and reference measurements. Third, outdoor field trials will be conducted at the test site, using the portable PV testbed at multiple locations, to assess SLAM robustness, point cloud quality, and arm positioning repeatability under realistic environmental conditions. Future software development will focus on three areas: improved surface specific filtering for reflective panel materials, semantic segmentation to enable class-specific anomaly thresholds, and integration of geometric observations with system level electrical data from the inverter-based testbed.

7 Conclusion

The paper describes the design and intended inspection workflow of a LiDAR-enabled ground rover for condition monitoring in utility-scale solar farms. The system addresses a practical gap: many structural and site-condition anomalies in large PV plants develop gradually and remain undetected by electrical monitoring and aerial inspection until they become more serious problems. A ground robot equipped with an articulated LiDAR arm can access viewpoints that drones cannot, return to the same position more reliably, and generate spatially dense geometric records that support change detection over time.

At this stage, the contribution is architectural and methodological. The platform has been assembled, the sensing stack is operational, and early laboratory trials have already informed a meaningful design decision by motivating the shift from a fixed LiDAR mount to an arm-mounted configuration. Field validation is the immediate next step, and the planned program is designed to characterize the performance of the system under realistic outdoor conditions before a broader deployment is considered.

In summary, as utility-scale solar farms continue to grow in size and age, the need for proactive and spatially informed O&M will become even more important. This work offers a practical foundation for developing that capability.

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